Diagram Design: AI-Powered Diagram Generation

Created by Cathryn Lavery (LittleMight) · Open-Source AI Agent Skill

Project Overview & Purpose OPEN SOURCE

Diagram Design is an open-source Agent Skill developed by designer and entrepreneur **Cathryn Lavery**. It empowers AI coding agents to generate publication-grade technical diagrams as single, self-contained HTML files with inline SVG—free of bloated libraries, heavy renderers, or external dependencies.

The Traditional Diagramming Problem PROBLEM SPACE

- **Manual Canvas Dragging:** Tools like Figma and draw.io require tedious manual alignment, box nudging, and color matching that disrupts developer focus.
- **Generic Automated Tools:** Standard text-to-diagram generators (e.g., default Mermaid or PlantUML) often produce visually unrefined layouts with excessive shadows, rigid auto-spacing, and non-branded themes.

Key Architectural Distinction: Diagram Design is *not* a standalone desktop graphical editor or an independent AI model. It is an **agent skill**—a structured knowledge framework that teaches host AI coding agents (Claude Code, Cursor, Codex) how to design with disciplined editorial taste.

GENERATION WORKFLOW ARCHITECTURE



CORE MEDIUM	GRAPHICS ENGINE	RUNTIME DEP
Self-Contained HTML	Inline Responsive SVG	Zero / Browser Native

How It Works

Four-Stage Operational Pipeline & Core Technical Capabilities

4-STAGE GENERATION PIPELINE

1

Describe

The user describes the target architecture, workflow, data model, or sequence using conversational, natural-language prompts.

2

Select & Plan

The AI agent identifies the best-fit diagram type from 45 specifications, loads layout budgets, and calculates coordinates on a strict 4px grid.

3

Generate

The agent writes clean, self-contained semantic HTML embedding responsive inline SVG, 1px hairlines, and accessible ARIA attributes.

4

Render & View

The user opens the resulting `.html` file immediately in any browser with zero compilation, dependencies, or external scripts.

01 45 Diagram Types

Comprehensive coverage: Architecture, Sequence, ER, State, Medallion, Sankey, Radar, Gantt, Axonometric, and more.

02 Natural-Language Creation

Full lifecycle from draft to refinement handled conversationally—request additions, changes, or rewiring in plain words.

03 Consistent Visual Styling

Disciplined editorial aesthetic: crisp 4px alignment, 1px hairlines, no drop shadows, and 1 focal accent color.

04 Brand Customisation

60-second onboarding pulls colors and typography from any website URL, verifying WCAG AA contrast automatically.

05 Import & Redesign

Ingests raw `.drawio`, Mermaid, and Excalidraw files, completely redrawing them into branded editorial diagrams.

06 Self-Contained Output

Single portable HTML files with inline SVG; zero external runtime dependencies, node packages, or rendering servers.

Important Architectural Note: Diagram Design does not train or host an AI model. It is an agent skill that equips general-purpose LLMs with visual taste, spatial constraints, and architectural templates.

Advantages of Diagram Design

Six Structural Benefits for Engineering Teams and Technical Communicators

Reduced Manual Effort

Automates tedious vector alignment, box sizing, and manual coordinate snapping, freeing developers from manual drag-and-drop canvas editing.

Consistent Visual Hierarchy

Enforces a disciplined 4px spatial grid, crisp 1px hairlines, and a single focal accent color to maintain clean editorial clarity across all outputs.

Natural-Language Creation

Enables users to describe and modify complex system topologies using everyday technical prose rather than intricate coordinate scripting.

Diverse Technical Scenarios

Covers 45 standardized diagram specifications spanning high-level architecture, low-level schemas, workflows, Sankey flows, and 3D axonometric plans.

Portable Browser Output

Produces self-contained HTML files with inline SVG that open instantly in any web browser without specialized viewer applications or proprietary software locks.

Cross-Agent Compatibility

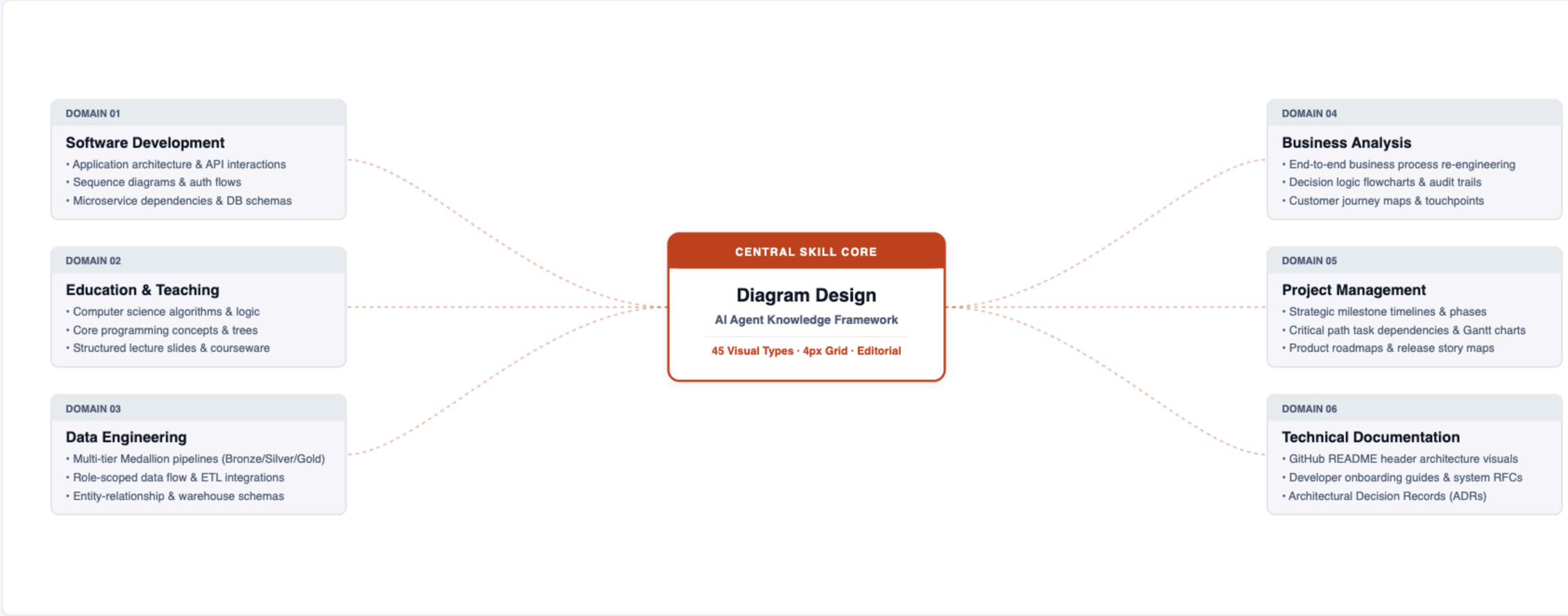
Operates uniformly across multiple supported AI coding tools including Cursor, Claude Code, Codex, GitHub Copilot, Qoder, and Factory Droid.

CORE TAKEAWAY **"Diagram Design helps developers communicate complex systems more clearly while reducing the effort required to create and style diagrams."**

Source: github.com/cathrynlavery/diagram-design

Where Can Diagram Design Be Used?

Six Technical Domains Powered by Structured Natural-Language Visualisation



PRACTICAL WALKTHROUGH

"A developer describes a web application's frontend, backend, database, and cache. The AI agent uses **Diagram Design** to create a structured architecture diagram with clean boundaries, precise 4px alignment, and zero design debt."

Source: github.com/cathrynlavery/diagram-design

PROJECT SYNTHESIS & EVOLUTIONARY ROADMAP

Future Scope and Conclusion

Distinguishing Production Features from Proposed Roadmap Additions

Existing Capabilities

PRODUCTION READY

45 Visual Diagram Types
Comprehensive reference library spanning architecture, data flows, sequences, and charts.

Automated Brand Onboarding
Scrapes URL palettes and fonts, verifying WCAG AA color contrast automatically.

Multi-Format Redesign & Import
Native parsing and editorial redrawing of raw draw.io, Excalidraw, and Mermaid inputs.

Zero-Dependency Browser Output
Self-contained HTML + SVG files with clean responsive rendering and optional headless export.

Proposed Future Additions

EXPLORATORY ROADMAP

ROADMAP 01
Drag-and-Drop Editing
An interactive browser canvas to manually nudge nodes, tweak alignment, or adjust coordinates post-generation.

ROADMAP 02
Real-Time Collaboration
Multi-user review threads, shared comments, and simultaneous live editing directly on diagram files.

ROADMAP 03
Multi-Format Native Export
Direct 1-click export to native PowerPoint (.pptx), vector PDF, and raster PNG without headless CLI tooling.

ROADMAP 04
Version History & Diffs
Semantic change tracking and visual diffs to audit additions, removals, and rewiring across Git commits.

ROADMAP 05
Enhanced Auto-Validation
Automated heuristics to flag disconnected components, dangling relationships, and overlapping bounding boxes.

ROADMAP 06
Expanded Agent Ecosystem
Broader integration across new web IDEs, autonomous documentation bots, and CI/CD diagram generators.

ACADEMIC & PROJECT CONCLUSION

"Diagram Design demonstrates how AI agent skills can turn natural-language instructions into structured, visually consistent diagrams. Its approach can benefit software development, education, data engineering, business analysis, and technical documentation."